* Unit - 2 *
Cellular Architecture

* Cellular Architecture is a wireless communication system in which large geographical area is divided into smaller are called cells. Each ~~serve~~ cell is served by a base station (BS) which provide wireless communication to mobile users within dead cell.

Multipath Access techniques

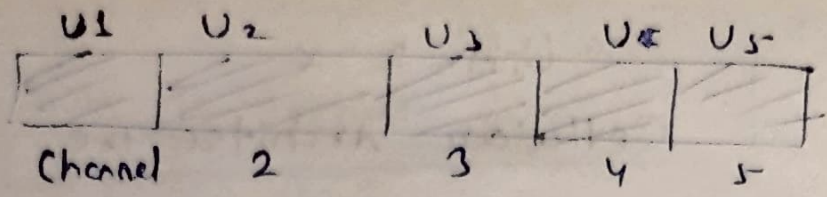
That allows multiple user/subscribers to simultaneously share the available radio resources of a communication system.

Types of Multiple Path Access Techniques -

1. FDMA
2. TDMA
3. CDMA

1) FDMA (Frequency Division Multiple Access)

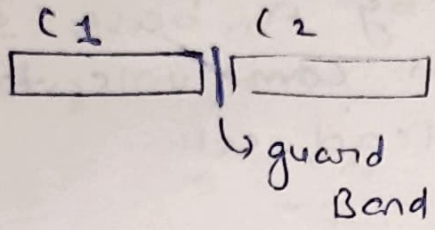
It is multiple path technique in which the available frequency spectrum is devided into saperate frequency channels & each user is assigned a different frequency channel.



Ex. FM radio frequencies.

$$f_1 \neq f_2 \neq f_3 \neq f_4$$

Two adjecent channel placed immediately next to each other



$$N_{FDMA} = \frac{BW_{total} - BW_{guard}}{BW_c}$$

Here BW_{guard} = Bandwidth of guard band.

BW_c = required bandwidth per channel.

Application

- In mobile Net
- Radio & TV broadcasting
- Satellite Communication

Advantage

+ Continuous transmission.
+ Less complex circuitry
+ Less interference.

Capicity

$$N = \frac{B_{total} - 2B_{guard}}{B_{channel}}$$

Disadvantage

+ Shared band waste some spectrum
+ Each user occupies a dedicated freq. channel
+ Costly.

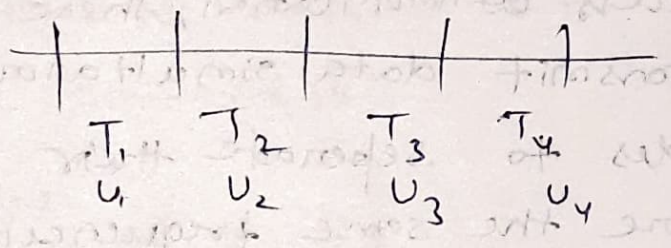
Time division Multiple Access (TDMA) :-

It is a channel access method used in wireless & digital communication systems who allows multiple user to share the same frequency channel by deviding the signal into different time slots.

Each user get the entire benifit but only for a short period of time.

∴ It reduce interference & effectively used Bandwidth.

→ available freq band is divided into time slots & different users are assigned different time slots.



It has guard time

U1 | GT | U2 | GT

Application

+ 2G Cellular Network
+ VSAT communication
+ Land mobile Radio & DMR.
+ Military Tactical data link

Advantage

+ Multiple uses can share the same carriers.
+ System capacity can increase
+ TDMA is digital
+ Cost can be lower.

Disadvantage:-

1. It requires precise synchronization
2. Time delay due to slot waiting.
3. Not ideal for high speed or continuous data.

Channel capacity in TDMA:-

$$R_{user} = R_{Total} / N$$

Here R_{total} = total data rate of the channel (bps)

N = Number of time slots (or users)

R_{user} = Data Rate per user.

* CDMA:-

In wireless communication, There multiple user can transmit data simultaneously using unique codes to seperate their signals. All user share the same frequency & time.

→ Each user data spread over using a unique code.

→ The receiver uses the same code to decode the intended data or signals & ignore others.

User A data – Code A – Spreader
User B data – Code B – Spreader — Combined Symbol
User C data – Code C – Spreader

At Receiver side

Combined Signal – Code A → Despreader – Receiver – User data.

Application

+ 3G mobile Network
+ Setellite communication.
+ In defence communication
+ In sensor networks

Advantage

+ High user capacity
+ Secure & interference Resistant
+ It makes better use of bandwidth.
+ As comparison to FDMA & TDMA
+ Security & privacy.

Disadvantage

+ It require complex hardware & software Installation at receiver site
+ Code limitations.

Capacity :-

$$N = \frac{G_p}{E_B / N_0}$$

where N = Number of users

$G_p = \frac{W}{R}$ = Processing Gain

(W = chiprate, R = Bitrate)

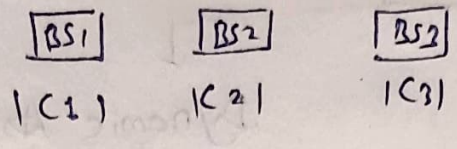
E_B / N_0 = Required energy per bit to noise Power density.

Need of Cells

for One base station working.

→ Limited Coverage
→ limited Capicity
→ High power requirement

So we need to work with cells.



→ Hexagon is used as an idealized model, as it can be placed next to each other without leaving gaps.

→ It provides complete coverage
→ Easier mathematical analysis
→ Six neighboring cells.

Capicity of Channels

$$C_{system} = M \times S$$

M = number of cluster
S = channel used at once.

or $C_{system} = C(S/N)$

here C = number of cell
N = cluster size
S = Total channel

Frequency Reuse factor = $1/N$

* Frequency Reuse In Wireless Communication:-

It is a technique used in cellular networks to maximize the ~~utilaction~~ utilazations of limited frequency spectrum. It involve reusing the same frequency channel in different geographical areas. Seperated by a sufficient ~~fastance~~ distance to avoid interference.

- **Why this frequency reuse important :-**
  - Spectrum is limited and expensive.
  - To support more users the same frequency need to be reused in different location.
  - Without frequency reuse each new need a separate frequency band which is impractical.

- **Frequency reuse distance**

$$D = R\,3\sqrt{N}$$

Ratio

$$Q = D/R$$

R = cell radius
D = distance
Q = frequency.

D = Distance between cells using the same frequency.
R = Radius of cell
N = Cluster size

$$Q = \sqrt{3N}$$

It create cluster that is group of cells in which all available frequency channels are assigned once according to a reuse pattern.

$$(N = j^2 + ij + j^2)$$

* Channel Assignment

Channel assignment is the process of allocating available radio channels to different cells to users in a cellular communication system.

Types of channel Assignment

- Fixed Channel Assigned (FCA)
- Dynamic Channel assignment. (DCA)

• Fixed Channel Assigment

Each cell is permanently allocated a specific set of channels.

Cell A	Cell B	cell C
(1 to 10)	(11 to 20)	(21 to 30)

Advantage

1. The channel allocation is predetermined
2. Easy to implement.
3. Lower conceptually complexity.

Disadvantage

1. Poor flexibility
2. Channel blocking can increase.

U1 U2 U3 U4 already assigned,
C1 C2 C3 C4

if any 5th user request
No free channel available.

- **Dynamic Channel Assignment (DCA)**

Channel are not permanently assigned to individual cells.

User requests call → System checks available channel.

↓

checks interference condition

↓

select a suitable channel → Channel assigned.

Advantage

1. Better channel utilization
2. Adapts to changing traffic
3. Can reduce blocking probability.

Disadvantage

1. More complex
2. More computation / signaling

* **Hand-off** :-

Process of transferring an ongoing call or communication connection from one cell/base station to another cell/base station as a mobile user moves through the cellular network.

→ It maintain continuous communication.

→ It handles prevents call interruption.

It detects → decide → Transfer → Then Release.

It occurs when
- Signal strength become weak
- Mobile moves from one cell to another
- A better base station becomes available
- The current cell becomes heavily loaded.

- **Handoff Threshold**

A cellular generally uses a threshold level.

P_r = received signal power.

when $P_r < P_{threashold}$

system initiate a handoff.

Types of Handoff → Hard handoff
→ Soft handoff

Hard handoff	Soft handoff
Break before make	Make before break
Old connection released first	old connection maintained temporarily.
New connection established after break	New connection established before old connection released.
High Risk of interruption	Low risk of interruption.

- **Handoff failure**

If the target cell does not have sufficient resources

Therefore, proper channel assignment & system capacity are closely connected with handoff.

* Interface & System Capacity:-

Interface is the disturbance caused by unwanted signals that affect the reception & quality of the desired signal.

So the result can be -

- Poor call quality
- Data errors
- Call drops,
- Reduced system performance.

Type of interference

↓ Co-Channel Interference

↓ Adjecent-channel Interference.

* Co-Channel Interference (CCI):-

Co-Channel interference is interference caused by signals transmitted using the same frequency channel in different cells.

So Same frequency -> Possible Co-channel Interfece.

Signal to interference Ratio (S/I)

$$\text{ratio} = S/I$$

S = desired signal power

I = Interference power.

High S/I → Better quality.

Co-Channel Reuse Ratio.

$$Q = D/R \qquad (D = \sqrt{3}N)$$

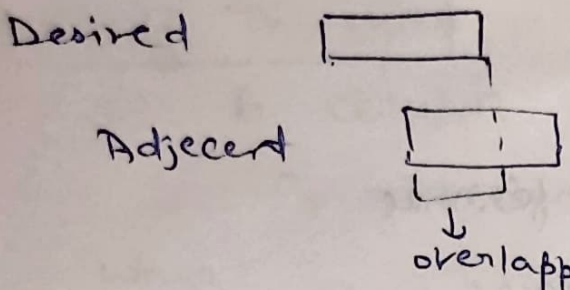
D = Distance bet centers of co-channel cell

R = Cell Radius

So N ↑ → Capacity ↓ → Interference ↓

- **Adjacent - Channel Interference (ACI)**

Adjacent - channel interference is interference caused by signals transmitted on frequency channels that are close to the desired channel.



Reduction of Adjacent - channel Interference

1. Proper channel Assignment
2. Use of guard band b/w channels gaps.
3. Better Receiver filtering -

* **Trunking & Grade of Service (GoS) :-**

Trunking is a technique in which a group of users shares a common pool of communication channels, & a channel is assigned to a user only when required.

without trunking

U1 → channel 1
U2 → channel 2
U3 → channel 3

with trunking

C1	C2	C3	C4	C5

↑
user request channel
↓
assigned only needed.

Grade of Service (GoS) is a measure of the ability of a communication system to provide service to users, commonly expressed in terms of the probability that a call will be blocked or delayed.

$$\text{Blocking probability} = \frac{\text{Blocked Calls}}{\text{Total call Attempts}}$$

Traffic intensity represents how much communication traffic is offered to a system.

* Erlang

$$A = \lambda H$$

- A: offered traffic intensity in erlangs
- λ: avg cell arrival rate
- H: avg call holding rate

1 erlang means, on avg one channel is continuously occupied during the observation period.

Trunking → sharing channel effeciently
GoS → measures how well the channel pool serves users.

* Coverage & Capacity Improvement :-

Coverage Problem → Some areas may receive weak signals.

Capacity problem → To many user try to use the same cell.

So we use Techniques to improve Capacity & Coverage
- cell splitting
- cell sectoring
- Repeaters.

1) Cell Splitting :-

Cell splitting is the process of dividing a large congested cell into several smaller cells to increase the capacity of the cellular system

Smaller cell → More Base station → More freq. reuse

↓

Higher capacity.

Cell Radius will reduces after splitting.

($R_{new} < R_{old}$)

Advantage

+ Increase system capacity
+ Handles high traffic areas.
+ Allow more freq. reuse
+ Improve service availability

Disadvantage

+ More base station required
+ Infrastructure cost increases
+ More handoff's may occur.

2) Cell Sectoring :-

Sectoring is the technique of dividing a cell into sectors using directional antennas in order to reduce interference & improve system capacity.

Directional Antenna → Signal concentrated in sector

↓

Interference reduced.

If interference ↓ then capacity ↑

3) Repeaters Used to improve coverage.

A repeater is a device that receives a weak signal, amplifies or regenerates it, & retransmits it to extend communication coverage

Base station ——→ weak coverage
signal become weak

Base station → Repeater → Improved Coverage

Coverage Improvement

Reach areas with weak or no signal

↳ - Repeaters
- Additional base station
- Better antenna pattern.

Capacity Improvement

Serve more simultaneous users

↳ - Cell spliting
- Sectoring
- More efficient freq. reuse.